

Even Examination Semester-2021

Course Code: PHYGCC03

Course Title: Electricity and Magnetism

Department of Appearing Students: PHYSICS, SEM-II, 1st Year

Full Marks: 40

Time: 2 hrs

Group-A

Answer any eight questions

8×5

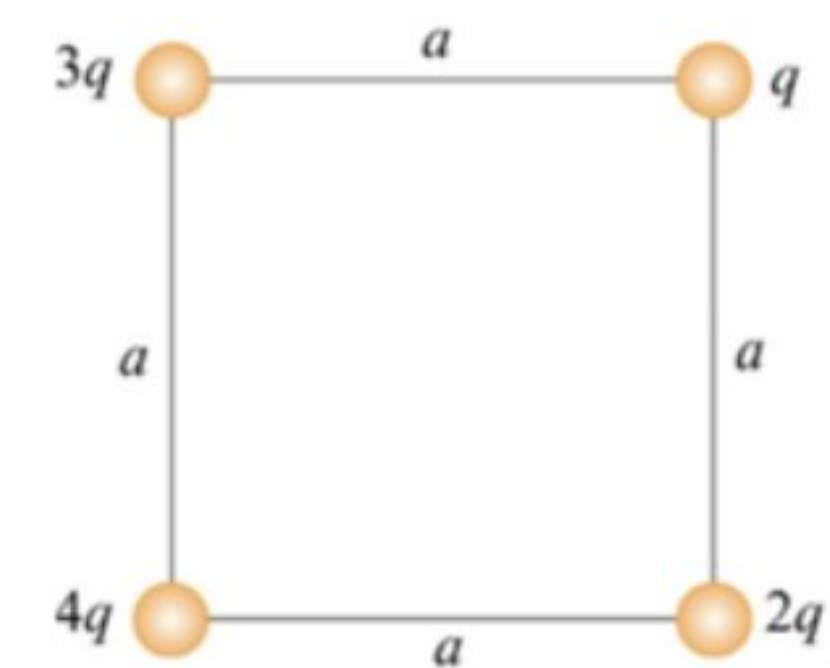
1. Explain Coulomb's law. Four point charges are placed at the corners of a square of side a , as shown in the figure below

(a) What is the electric field at the location of charge q ?

(b) What is the net force on $2q$?

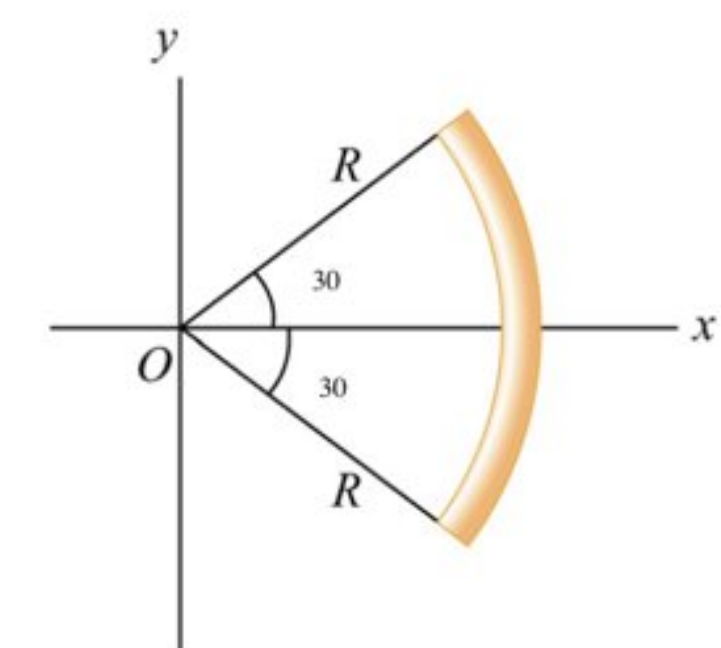
(In both cases sketch the individual and resultant forces on each charge.)

1+2+2



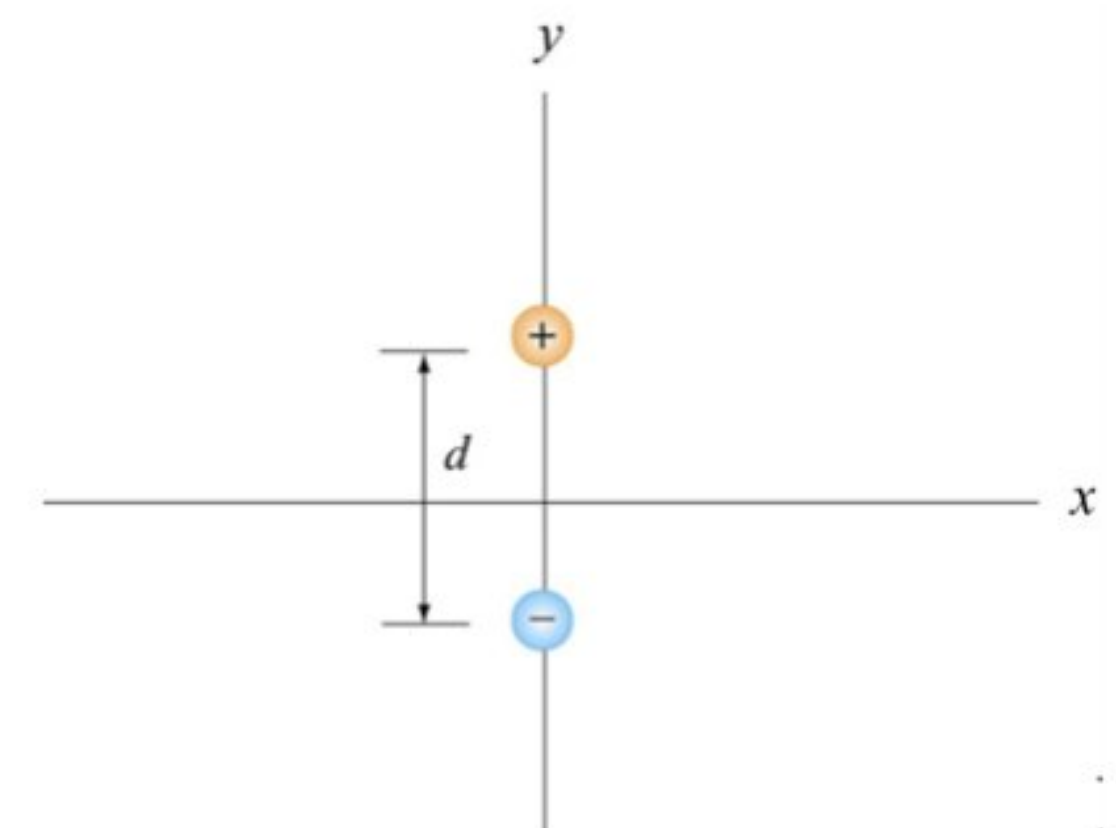
2. A thin rod with a uniform charge per unit length λ is bent into the shape of an arc of a circle of radius R . The arc subtends a total angle 60° , symmetric about the x -axis, as shown in the figure below. What is the electric field $\mathbf{E}$ at the origin O ?

5



3. What is electric dipole? Deduce the amount of torque on a dipole with moment $\mathbf{p}$.

An electric dipole consists of two charges $q_1 = +2e$ and $q_2 = -2e$ ($e = 1.6 \times 10^{-19}$ C), separated by a distance $d = 10^{-9}$ m. The electric charges are placed along the y -axis as shown in Figure. Suppose a constant external electric field $\vec{E} = (3\hat{i} + 3\hat{j})$ N/C is applied. What is the magnitude and direction of the dipole moment and the torque on the dipole?

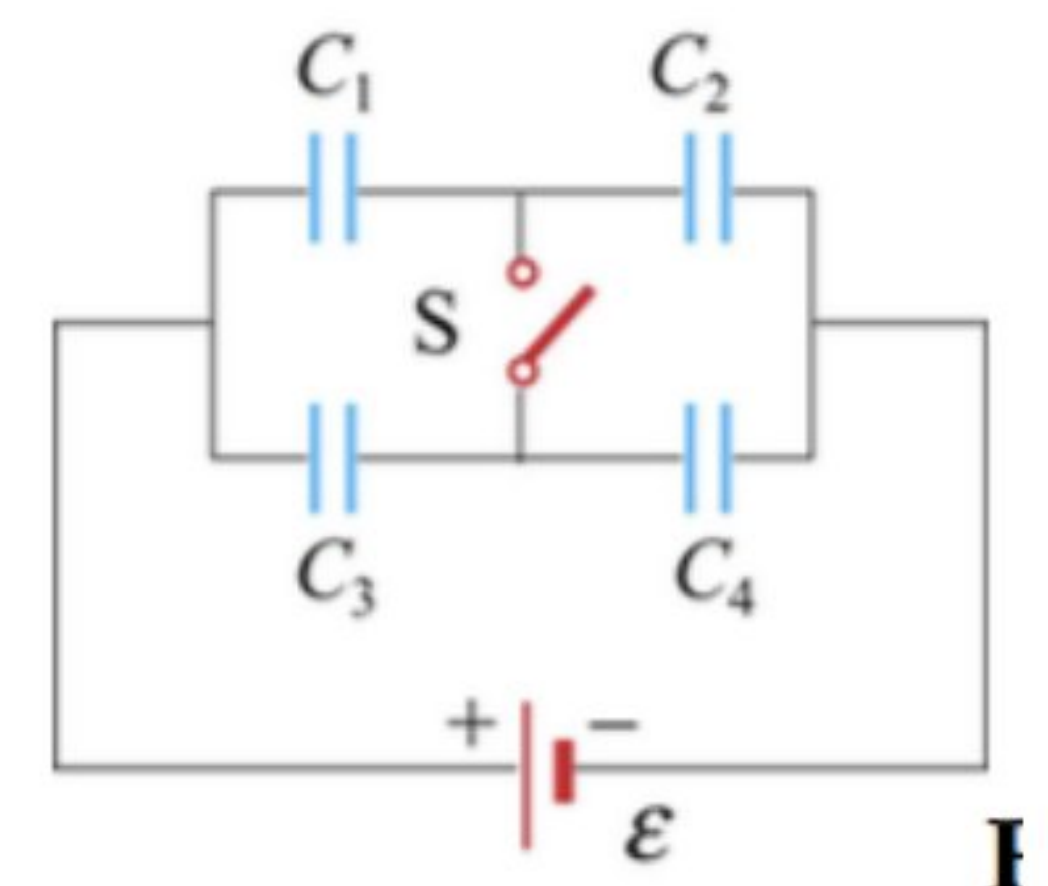


4. A 12V battery charges the four capacitors as shown in the Figure. Let $C_1 = 1\text{F}$, $C_2 = 2\text{F}$, $C_3 = 3\text{F}$ and $C_4 = 4\text{F}$

(a) What is the equivalent capacitance of the group C_1 and C_2 if the switch S is open?

(b) What is the charge on each of the four capacitors if switch S is open?

3+2



5. State Biot Savart and Ampere's circuital law.
Find the magnetic field at a point P on the axis of the current carrying loop, at a distance z from the center?
6. Define vector potential for a magnetic field. Show that a possible choice of the vector potential for a constant magnetic field $\vec{B}$ is given by $\vec{A} = \left(\frac{1}{2}\right)(\vec{B} \times \vec{r})$.
7. A series RC circuit with $R = 4.0 \times 10^3$ and $C = 0.4 \mu F$ is connected to an AC voltage source $v(t) = 100 \sin \omega t$, with $\omega = 200 \text{ rad/s}$
 - (a) What is the rms current in the circuit?
 - (b) What is the phase difference between the voltage and the current?
 - (c) Find the power dissipated in the circuit.
 - (d) Find the voltage drop both across the resistor and the capacitor. 5
8. In an RC series circuit, $E = 12.0 \text{ V}$, $R = 1.40 \text{ M}\Omega$, and $C = 1.80 \mu F$.
 - (a) Calculate the time constant.
 - (b) Find the maximum charge that will appear on the capacitor during charging.
 - (c) How long does it take for the charge to build up to $16.0 \mu C$? 5
9. In a LR circuit a DC voltage V is applied. Deduce the expression for transient current. What is steady state? Find the transient current if the voltage is switched off. 3+2
10. (a) Capacitance of parallel plate capacitor becomes $4/3$ times its original value if a dielectric slab of thickness $t = d/2$ (d is the separation between the plates) is inserted between the plates. Find the dielectric constant of the.
 (b) A long string with a charge of λ per unit length passes through an imaginary cube of edge l . Find the maximum possible flux of the electrical field through the cube. 2.5+2.5
11. Discuss Gauss's law of electrostatics. Using Gauss's law find the electric field at a inside and outside point of hollow sphere carrying charge Q . 2+3

Semester Examination -2021

Course Code: PHYUGCC04

Course Title: Waves and Optics

Department of Appearing Students: PHYSICS, II SEM, 1st Year

Full Marks: 40

Time: 2 hrs

Answer any four

1. a) What is the condition for constructive and destructive interference?
 b) Maximum intensity in Young's double slit experiment is I_0 . Find the intensity at a point on the screen where i) the phase difference between the two interfering beam is $\pi/3$ ii) the path difference between them is $\lambda/4$
 c) Calculate the minimum thickness of a soap film ($\mu = 1.33$) that results in constructive interference in the reflected light if the film is illuminating with light whose wavelength in free space is $\lambda = 600nm$.

2+5+3

2. a) Newton's rings are observed in the reflected system with wavelength $5.9 \times 10^{-5} cm$. If diameter of 8th dark ring is 0.5 cm, find the radius of curvature of the lens and thickness of air film.
 b) What is the main difference between Fraunhofer diffraction and Fresnel diffraction?
 c) What is the highest order spectrum, which may be seen with monochromatic light of wavelength $6000 \text{ \AA}$ by means of a diffraction grating with 5000 lines/cm?
 d) What is the radius of the first zone in a zone plate of focal length 20cm for light of wavelength $5000 \text{ \AA}$?

3+2+3+2

3. a) Show that the intensity distribution for the diffraction in a single slit is given by $I = I_0 \frac{\sin^2 \beta}{\beta^2}$, where the symbols have their usual meaning. Deduce the condition for maxima and minima from it.
 b) A light of wavelength $5893 \text{ \AA}$ is passes through a slit, which is 0.5 mm wide and forms a diffraction pattern on a screen placed on the focal plane of a lens of focal length 1.0m. Calculate the separation of the dark band on either side of the central maximum.
 c) Calculate the minimum number of lines in a grating that will first resolve the lines of wavelengths $5890 \text{ \AA}$ and $5896 \text{ \AA}$ in the second order.

(3+2)+3+2

4. a) Calculate the resultant of two SHMs at right angles when their periods are at the ratio 1:2 and there is a phase difference 0 or $\pi/2$.

(b) Show that centre of suspension and centre of oscillation are interchangeable for compound pendulum.

© Show that the energy density of progressive wave is proportional to the density of the medium but proportional to the square of amplitude and frequency.

4+3+3

5. a) Obtain the expression for the displacement in a damped oscillatory motion and discuss the of critical damping.

b) An object of mass 0.001kg hangs from a spring of spring constant 1000N/m. A resistive force rv acts on it where v is the velocity in m/s and $r=10\text{kg/s}$. If the object be subjected to a driving force $F=F_0 \cos(pt)$ with $F_0=20\text{N}$ and $p=5000 \text{ rad/s}$, then calculate the amplitude of oscillation and phase relative to force in the steady state.

(3+3)+4